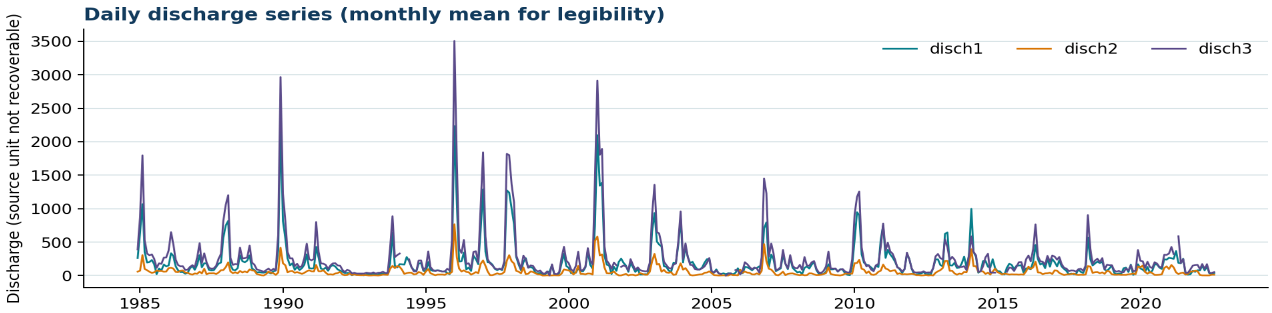
Use Case 1 - SNIRH River-Flow Forecasting

Multivariate daily forecasting for an environmental-monitoring IoT/CPS scenario

Dataset identity and quality

Checked CSV	use_case_1_river_flow_observed.csv
Source	Portuguese SNIRH portal
Rows / variables	13,715 observed timestamps / disch1, disch2, disch3
Covered period	1984-12-30 to 2022-08-06
Cadence	Daily, midnight observations
Calendar gaps	19 absent daily timestamps (13,734 timestamps on a complete index)
Missing cells	disch1: 170, disch2: 197, disch3: 958
Duplicates / order	0 duplicates; strictly increasing: True



Quality warning: six non-daily test rows dated 9 July 2025 and three empty columns were present in the supplied file and were excluded. A separate thesis consistency issue remains: the narrative identifies the third series as target, while the inspected executable listing declares output_features disch2. Resolve that target choice before claiming exact reproducibility.

Use-case role

The data support the thesis's SNIRH-based river-flow scenario. LSTM and GRU configurations use a 20-day multivariate input window and produce three daily forecast horizons. Forecast values are intended to feed a flood-monitoring component.

Execution-aligned preparation

The checked CSV retains observed midnight records before 7 August 2022. A complete daily index over that period contains 13,734 timestamps, so 19 calendar timestamps are absent from the supplied observations. The ML2++ preprocessing specification uses time interpolation only for internal gaps of at most ten consecutive days and constructs windows only inside valid common periods. Random shuffling and feature scaling are not used.

The thesis reports chronological partitions of 8,179 training samples, 2,045 validation samples, and 2,557 held-out test samples after preprocessing and window construction. Those derived sample counts are not raw-row counts.

Models and outputs

Models: LSTM and GRU. Inputs: disch1, disch2, and disch3. Look-back: 20 daily observations. Horizon: t+1, t+2, and t+3 days. Evaluation: horizon-specific RMSE, MAE, and R-squared as documented in the thesis.

Provenance note

The intended scenario mentions Albufeira da Aguieira (11H/01A), Albufeira da Raiva (12H/01A), Albufeira de Fronhas (12I/01A), and Acude Ponte de Coimbra (12G/01AE). However, the processed file does not contain station codes. Therefore, no disch-column-to-station mapping or measurement unit is claimed in this artefact.